

Seohyeon Yoo

HCI Researcher · Digital Healthcare

Phone: +82 10 9076-9854

Email: seohyeon.yoo@kaist.ac.kr

<http://www.linkedin.com/in/seohyeon-yoo>

EDUCATION

KAIST (Korea Advanced Institute of Science and Technology)

M.S. in AI Systems, Advisor: Sung-ju Lee

Sep.2026 – Present

Daejeon, Republic of Korea

Hanyang University (HYU)

B.S. in Data Science

Mar.2021 – Aug.2026

Seoul, Republic of Korea

Universidad Carlos III de Madrid (UC3M)

B.S. in Data Science (Exchange student)

Sep.2024 – Dec.2024

Madrid, Spain

PUBLICATIONS

Preprints

[p.1] Collective Voice: Recovered-Peer Support Mediated by An LLM-Based Chatbot for Eating Disorder Recovery

Ryuhaerang Choi, Taehan Kim*, Subin Park*, **Seohyeon Yoo**, Jennifer G Kim, and Sung-Ju Lee (* equal contribution.)

arXiv preprint

Conference Papers

[c.1] Recovery is Relational: Digital Support Needs for Patients and Supporters in Eating Disorder Recovery

Ryuhaerang Choi, **Seohyeon Yoo**, Xuhai Xu, and Sung-Ju Lee

CHI 2026: ACM CHI Conference on Human Factors in Computing Systems

Posters/Workshop Papers

[po.1] “Call Me Before I Open the Fridge”: Designing Conversational Support for Binge Eating Prevention During Solitary Eating

Ryuhaerang Choi, **Seohyeon Yoo**, and Sung-Ju Lee

UbiComp/ISWC 2026 Posters: ACM International Joint Conference on Pervasive and Ubiquitous Computing

WORK EXPERIENCE

Samsung Electronics DS Division (AI Center)

Engineering Intern

Sep.2025 – Dec.2025

Dongtan, Republic of Korea

- Designed and developed an unsupervised anomaly detection pipeline for semiconductor process data, automating preprocessing and reducing manual inspection through LLM-based validation.
- Built an LLM-based proposal review agent for internal AI consulting workflows, improving review consistency, output reliability, and hallucination control.

KAIST Mobile Intelligence & Interaction Lab (MIIL)

Research Intern (Advisor: Prof. Sung-Ju Lee)

Jul.2024 –Present

Daejeon, Republic of Korea

- Conducting research on early detection of binge eating episodes using behavioral and physiological data. (Ongoing)
- Investigated stakeholder perspectives on digital interventions for eating disorder recovery through user studies and mobile app deployment. [c.1]
- Examined how recovered-peer personas shape emotional resonance and meaning-making in an LLM-based recovery-support chatbot through qualitative interview coding. [p.1]

Hanyang Data Mining & Algorithm Laboratory*Researcher Intern (Advisor: Prof. Hyun-joon Kim)**Mar.2023 – Mar.2024**Seoul, Republic of Korea*

- Proposed a sequential recommendation framework improving next-item prediction under sparse and popularity-biased data via frequency-aware embeddings, and retrieval-based disentanglement.

PROJECTS

Coursework**Debiased Retrieval-Augmented Sequential Recommendation***Dec.2023 – Jun.2024*

- Investigated retrieval-augmented sequential recommendation for mitigating data sparsity and popularity bias.

Personalized Fitness Social Network Application*Oct.2023 – Dec.2023*

- Designed a mobile fitness social networking application as part of HCI coursework, incorporating personalization, gamification, and privacy-aware design.

External Projects**LLM-assisted digital emotional support web service***Mar.2025 – Jun.2025**Tech for Impact with KakaoImpact Foundation*

- Developed an LLM-assisted digital emotional support web service for adolescents experiencing elevated depressive symptoms, using guided letter-writing prompts to foster emotional expression and recovery. [c.2]

Interactive Art Education Platform for Children*Jan.2024 – Feb.2024**Google Developer Student Clubs*

- Developed an interactive art education platform for children from low-income families, focusing on accessibility and inclusive learning design.

Terrain Shape Classification Using Spatial Data*Jan.2023 – Feb.2023* *Spatial Information Utilization Contest*

- Developed a terrain shape classification model using CNN-based image representations of spatial data.

AWARDS. AND SCHOLARSHIP

Hanyang Changemaker Award*Feb.2026**Awarded by Hanyang University's Global Social Innovation Division***Gold Prize, Summer Undergraduate Paper Competition in DCS***Jul.2025**Hosted by Korea Digital Contents Society***Bronze Prize, Summer Undergraduate Paper Competition in KIIT***Jun.2025**Hosted by Korea Institute of Information Technology***Innovative Technology Award, Techno Impact with Kakao***Jun.2025**Hosted by Kakao Impact Foundation and Hanyang University***Undergraduate Leadership Scholarship***Dec.2023**Awarded for service as the student representative of the Data Science Department, 3rd Year.***Hanyang Brain (Academic Excellence) Scholarship***Jun.2023**Awarded to a student for academic excellence in the Data Science Department throughout the semester***2nd place, Spatial Information Utilization Contest***Feb.2023**Hosted by the spatial information research institute of Korea land and geospatial informatiX corporation (LX)***Student Media Production Team Scholarship, Hanyang University***Dec.2022***Undergraduate Leadership Scholarship***Dec.2022**Awarded for service as a Team Leader in the Student Council of the College of Engineering*

ACADEMIC SERVICE

Reviewer

ACM CHI 2026 Posters — Special Recognition for Outstanding Reviews

ACM CUI 2026 Short Papers & Works-in-Progress

ACM UbiComp/ISWC 2026 Posters

ACM MobileHCI 2026 Late-Breaking Work

EXTRACURRICULAR ACTIVITY

LG Aimers 7th	<i>Jul.2025 – Aug.2025</i>
---------------------------------	----------------------------

Tech for Impact with KakaoImpact Foundation	<i>Mar.2025 – Jun.2025</i>
--	----------------------------

Google Developer Student Clubs (GDSC)	<i>Sep.2023 – Jun.2024</i>
--	----------------------------

Hanyang University Student Media Production Team(HY-IM)	<i>Mar.2022 – Feb.2023</i>
--	----------------------------

College of Engineering Student Council	<i>Mar.2022 – Dec.2022</i>
---	----------------------------

Department of Data Science Student Council	<i>Mar.2021 – Dec.2022</i>
---	----------------------------

KEY SKILLS

Programming: Python, Typescript

Research Methods: User study, Prototyping, Qualitative data coding, Quantitative analysis

Others

English Proficiency: TEPS (364), TOEIC (935)

Certification: ADsP